

ITU Noticeboard Patrol: Autonomous Observation and Anomaly Detection

YZV406E Robotics — Project Proposal (Fall 2025)

Team

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Project Topic

Design and simulate a mobile robot that **patrols university corridors, approaches noticeboards, captures high-resolution poster images, and performs information extraction and analysis** to detect *expired or outdated* announcements and generate structured reports.

Problem Definition & Motivation

Highly used buildings (e.g., Architecture, EEF) contain many noticeboards whose content changes frequently. Manually checking for *expired* or *duplicate* announcements is tedious. We propose an autonomous system that safely navigates crowded corridors, aligns to boards, reads posters, flags anomalies (e.g., past-due dates), and logs results for facility staff.

Environment & Platform

- **Simulation:** Gazebo Harmonic indoor corridor with multiple noticeboards, benches, moving pedestrians.
- **Robot platform:** TurtleBot-class differential drive robot equipped with a 2D LiDAR for mapping and obstacle avoidance, and an RGB or RGB-D camera for board detection and image capture. ROS2 Humble will be used.

High-Level Requirements

1. **Safe Navigation & Approach:** Build or obtain a 2D map, localize, avoid pedestrians, and *dock* to a board viewing pose.
2. **Perception & Information Processing:** Identify and localize noticeboards, capture images, and process visual data to extract relevant content such as titles, dates, and general summaries. Detect potential anomalies including outdated or duplicated announcements. Methods may involve image-based detection, fiducial markers or external services.
3. **Data Logging:** Record board IDs, timestamps, captured images, extracted fields, and anomaly flags for reporting and traceability.
4. **Testing & Metrics:** Evaluate success rate, navigation safety, approach accuracy, perception robustness, and information analysis precision/recall, including demonstration through recorded video presentations.

Method & Approach

Perception (Boards and Posters)

- **Board and poster observation:** The robot identifies noticeboards, aligns its view, and captures high-quality images ensuring sufficient framing, resolution, and lighting.
- **Information extraction and analysis:** Captured visual data is processed or transmitted to obtain relevant content such as titles, dates, or summaries. Potentially outdated or duplicated announcements are flagged. The method may involve onboard processing or external services.
- **Optional refinement:** When confidence in extracted information is low, secondary reasoning or external interpretation modules may be applied to improve accuracy.

Localization and Mapping

- The robot constructs an internal map of the environment, identifying corridors and board positions.
- During operation, it continuously updates its own position on this map, allowing it to navigate accurately and approach each board safely.

Planning and Control

- The robot plans efficient and safe routes between all target boards, considering environmental constraints such as corridor width and pedestrian movement.
- Motion control adjusts velocity and direction dynamically to avoid collisions and maintain smooth navigation.
- Upon reaching a board, the robot stops at a suitable viewing distance and adjusts its orientation to face the board frontally.
- The full behavior sequence follows a structured routine: *navigate to board* → *align view* → *capture image* → *extract and analyze text* → *log findings* → *proceed to next board*.

Individual Responsibilities (Distinct Robotic Subtasks)

Member	Primary Task	Sample Subtasks and Duties
Metin Ertekin Küçük	Perception	Detection and localization of noticeboards; capturing and processing poster images to extract relevant information; identifying expired or duplicated announcements. Video duties.
Abdullah Saatçi	Navigation and Control	Developing safe movement strategies in indoor corridors; avoiding dynamic obstacles such as pedestrians; designing and testing smooth approach behavior to stop at appropriate viewing distance from boards. Presentation duties.
Emre Saraç	High-Level Planning and Testing	Defining overall patrol routes and visit sequences; coordinating perception and motion behaviors; running integrated system tests and analyzing performance across different crowd and lighting conditions.

Timeline (Dates reflect course call)

Milestone	Target Dates	Details
Project start	5–11 Nov 2025	Repository setup, simulation environment preparation, base robot configuration.
Proposal submission	12 Nov 2025	Finalize topic, goals, and team task distribution.
Mapping and movement baseline	13–19 Nov 2025	Establish initial map of corridors and test basic navigation routes.
Perception prototype	20–26 Nov 2025	Implement board detection and image capture on simulated noticeboards.
Integrated patrol behavior	27 Nov–3 Dec 2025	Combine perception and motion behaviors into a complete patrol cycle.
Testing and tuning	4–10 Dec 2025	Evaluate navigation safety and perception performance under varying crowd density.
Polish and documentation	11–17 Dec 2025	Final adjustments, report writing, and demo preparation.
Final demonstration	18–29 Dec 2025	Conduct full scenario run and present results.